# ARIEL RENEWABLE ENERGY LIMITED

## Customer Segmentation, Commercial Packages, Technical Specifications, Costing & Pitch Masterclass

**Corporate Entity:** Ariel Renewable Energy Limited
**Parent Group:** ArielMC
**Business Sector:** Energy
**Product:** Solar (Turnkey Rooftop Solar EPC)
**Core Web Portal:** energy.arielmc.com
**Command Hub:** energy.arielmc.com/crm

## SECTION A: COMMERCIAL PACKAGES MATRIX (3 CUSTOMER CATEGORIES)

Ariel Renewable Energy structures its market entry across 3 distinct tiers: **Factory Industrial Sheds** (Mega C&I;), **Commercial Buildings** (Mid-scale), and **Households** (Residential & Societies).

### 1. Category 1: Factory & Industrial Export Manufacturers (300 kWp – 5,000 kWp)

| Package Name | Target Scale | Core Engineering Features | Financing & Payback |
|---|---|---|---|
| **Sovereign Export Prime (Turnkey Grid-Tied)** | 300 kWp to 2,500 kWp | Zero-penetration standing seam PEB clamps; SREDA 11kV/33kV net-metering; Accord/RSC 1.2m fire safety walkways; Jinko TOPCon 585W bifacial. | 80% Bangladesh Bank 4.5% Green Refinance; 0% working capital; Payback: **3.2 – 3.6 Years**. |
| **Industrial Hybrid Resilient (Solar + BESS Storage)** | 500 kWp Solar + 500kWh–2MWh BESS | Sub-20ms UPS switchover; Peak-hour tariff shaving (5-11 PM); Slashes diesel generator run time by up to 75%; CATL/BYD LiFePO4. | Concessionary green debt structuring; High fuel savings hedge; Payback: **3.8 – 4.4 Years**. |
| **LEED Carbon-Shield (I-REC Accelerator)** | 500 kWp to 5,000 kWp | Revenue-grade automated energy accounting; Blockchain-verified I-REC certificate issuance; EU CBAM compliance audit readiness. | Turnkey EPC + 10-Yr O&M; Guarantee; Top buyer priority status; Payback: **3.1 – 3.5 Years**. |

### 2. Category 2: Commercial Buildings & Corporate Centers (30 kWp – 300 kWp)

| Package Name | Target Scale | Core Engineering Features | Benefits & Payback |
|---|---|---|---|
| **Corporate Prestige (Elevated Pergola)** | 30 kWp to 150 kWp | 8-foot elevated hot-dip galvanized structural canopy; **100% roof space preserved** for executive gardens/cafes; Double-glass bifacial panels. | High aesthetic corporate green asset; Slashes building common bill 45-60%; Payback: **3.5 – 3.9 Years**. |
| **Retail & Mall Peak Shaver (Chiller Load Direct)** | 100 kWp to 400 kWp | Direct counter to high daytime AC chiller loads; Replaces high commercial tariff (Tk 13.50-15.00); Multi-MPPT Huawei/Sungrow inverters. | **Fastest payback in Bangladesh**: **3.0 – 3.4 Years**; Immediate monthly cashflow surplus. |
| **Healthcare Hybrid (Critical Continuity)** | 50 kWp to 250 kWp + BESS | Medical-grade power stability; Instantaneous battery backup for ICUs and surgical wards; Automatic diesel generator bridging. | Life-support reliability; Zero flicker; Hedge against grid brownouts; Payback: **3.9 – 4.5 Years**. |

## 3. Category 3: Households & Residential Communities (3 kWp – 30 kWp)

| Package Name | Target Scale | Core Engineering Features | Family & Society Value |
|---|---|---|---|
| **Urban Home Freedom (Duplex / Private Villa)** | 3 kWp to 6 kWp | Eliminates high residential tariff slabs (Tk 12–14.50/unit); Smart hybrid inverter with phone app; Compact footprint (250–450 sq.ft). | Slashes family monthly bill up to 80%; Free AC daytime power; Payback: **4.0 – 4.5 Years**. |
| **Apartment Common-Load (Shared Utility Sovereign)** | 10 kWp to 30 kWp | Powers common utilities: Lifts, water pumps, stairwell lighting, CCTV security; Non-penetrative ballast mounts protecting roof waterproofing. | **Permanently slashes flat service charge 40-55%**; Eliminates society budget friction; Payback: **3.8 – 4.2 Years**. |
| **Solar Gazebo Canopy (Garden Rooftop Pergola)** | 5 kWp to 12 kWp | Elevated 9-foot architectural pergola; Double-glass bifacial modules creating comfortable shaded outdoor tea lounge/patio below. | Protects rooftop garden from scorching heat; Beautiful luxury living extension; Payback: **4.2 – 4.8 Years**. |

## SECTION B: DETAILED SPECIFICATIONS & MULTI-OPTION HARDWARE CHOICES

Clients choose between Tier-1 Flagship and Value-Engineered configurations, backed by Ariel's 25-Year System Performance Assurance.

| System Component | Option 1: Tier-1 Flagship | Option 2: Value-Engineered | Warranties & Compliance |
|---|---|---|---|
| **Photovoltaic Modules** | **JinkoSolar Tiger Neo N-Type**<br>585W–600W TOPCon Bifacial<br>22.8% Eff, -0.30%/°C Temp Coeff | **LONGi Solar Hi-MO X6 / 7**<br>580W–595W HPBC / TOPCon<br>Dual-glass, Class A Fire Rating | 12-Yr Product Guarantee<br>30-Yr Linear Power Output (≥87.4%)<br>Degradation: ≤0.40%/yr |
| **Inverters & Electronics** | **Huawei SUN2000 Series**<br>50KTL to 115KTL-M2<br>AI Arc Fault Interrupter (AFCI) | **Sungrow SG Series**<br>SG110CX / SG125HX<br>150% DC/AC overload ratio | 5-Yr Standard / 10-Yr Ext Warranty<br>IP66 Outdoor Rated<br>Smart I-V Curve Diagnosis |
| **Mounting Substructures** | **Anodized Aluminum AL6005-T5**<br>Zero-penetration Klip-Lok clamps<br>SUS304 Stainless Fasteners | **Hot-Dip Galvanized Steel**<br>IS 2062 / ASTM A36 (≥80µm zinc)<br>Elevated Pergola 8–9 ft | 25-Yr Anti-Corrosion Warranty<br>Wind Resistance: Up to 210 km/h<br>**100% Zero Roof Holes Guarantee** |
| **Switchgear & Safety** | Schneider / ABB / Siemens MCCB<br>Type 1+2 DC/AC Surge Arresters<br>TUV 1500V XLPO Solar Cables | Chint / Suntree IP65 Combiners<br>Type II Surge Protection<br>Halogen-Free Flame Retardant | Dedicated Chemical Earth Pits<br>**Resistance < 0.8 Ohm**<br>Accord/RSC 1.2m Walkways |
| **IoT & Net-Metering** | **Ariel-SaaS™ 4G/LTE Gateway**<br>Per-string telemetry, cloud alerts<br>SREDA Bi-Directional Class 0.2S | Smart Logger + Modbus RTU<br>Mobile App dashboard<br>Class 0.5S Net-Meter | Real-time Revenue Metering<br>Automated SREDA monthly report<br>I-REC Carbon Ledger |

## SECTION C: COSTING WITH DETAIL PRICE BREAKDOWN & FINANCIAL ROI

Turnkey pricing benchmarks in Bangladesh Taka (BDT) and US Dollars, incorporating 1% fiscal customs duty and zero-tax status:

| Segment & Capacity | Cost per Watt (Wp) | Total Turnkey CapEx (BDT) | Annual Electricity Savings | Simple Payback |
|---|---|---|---|---|
| **Industrial 500 kWp** | ■ 52 / Wp ($0.43) | ■ 2,60,00,000 (Tk 2.60 Cr) | ■ 82,30,000 / Year | **3.16 Years** |
| **Industrial 1,000 kWp (1MW)** | ■ 50 / Wp ($0.42) | ■ 5,00,00,000 (Tk 5.00 Cr) | ■ 1,64,50,000 / Year | **3.04 Years** |
| **Commercial 50 kWp** | ■ 62 / Wp ($0.52) | ■ 31,00,000 (Tk 31.0 Lakh) | ■ 9,80,000 / Year | **3.16 Years** |
| **Commercial 100 kWp** | ■ 58 / Wp ($0.48) | ■ 58,00,000 (Tk 58.0 Lakh) | ■ 19,60,000 / Year | **2.96 Years** |
| **Residential 5 kWp Duplex** | ■ 72 / Wp ($0.60) | ■ 3,60,000 (Tk 3.60 Lakh) | ■ 88,750 / Year | **4.05 Years** |
| **Society 10 kWp Common** | ■ 68 / Wp ($0.57) | ■ 6,80,000 (Tk 6.80 Lakh) | ■ 1,75,000 / Year | **3.88 Years** |

**Standard Bill of Quantities (BOQ) Percentage Allocation:** PV Modules: 48% | Inverters: 18% | Mounting Structures: 11% | DC/AC Cabling & Protection: 9% | SREDA Licensing & Grid Sync: 6% | Installation & Rigging: 5% | IoT & Commissioning: 3%.

# SECTION D: CUSTOMER PITCH MASTERCLASS — PSYCHOLOGICAL FRAMING

**Core Directive:** Never pitch solar as a supplier selling goods. Frame the conversation around the *client's urgent financial self-interest*, making them realize *they are the primary beneficiary* while discharging a *noble social and environmental responsibility*.

## 1. The Factory Owner & Managing Director Pitch (CFO & MD Perspective)

*'Transforming an Idle Roof into Your Factory's Highest-Yielding Productive Asset'*

**Opening Script:**
"Mr. Managing Director, every square foot of your factory floor is engineered for peak output — except your roof. Right now, your 80,000 sq.ft roof is an idle asset absorbing scorching heat, raising interior temperatures, while your finance department pays millions of Taka every month to the power utility.

We are not here to ask for your working capital. Bangladesh Bank's 4.5% Green Refinancing Facility and SREDA Net-Metering allow your roof to pay for itself. Your monthly electricity savings will exceed your bank installment from Month 1, creating immediate positive cashflow. By Year 3.5, the installation is fully paid off — and for the next 21+ years, your factory enjoys **100% free electricity**.

With European buyers enforcing strict ESG and the EU Carbon Border Adjustment Mechanism (CBAM) taking effect, this installation guarantees your facility stands at the top of the buyer order list. You are not buying solar panels; you are securing your factory's global export future while leaving a cleaner Bangladesh for the next generation."

## 2. The Commercial Building & Corporate Tower Pitch (Asset & Facility Perspective)

*'The Double-Dividends of Architecture and Operating Expense Shield'*

**Opening Script:**
"Distinguished Chairman, commercial electricity tariffs are among the heaviest recurring burdens on any premier building in Dhaka. Every facility manager knows that daytime AC chillers drive utility bills to peak levels.

Ariel Renewable Energy has engineered a solution that requires zero compromise: our **Elevated Solar Pergola Architecture**. We do not take away your roof space. We elevate double-glass solar modules 8 to 9 feet on architectural galvanized steel, creating a cooler, shaded terrace garden for executive meetings, breakout events, and open-air hospitality — while the canopy above silently slashes your building's electricity bills by 45% to 60%.

Your commercial tower instantly earns a Green Building rating, commands 15–20% higher rental premiums from multinational tenants, and pays off the entire investment in just three years. This is not an expenditure; it is an intelligent asset enhancement that adds permanent capital value to your property."

## 3. The Homeowner & Residential Society Pitch (Family & Community Harmony)

*'Energy Sovereignty for Your Family, Harmony for Your Community'*

**For the Private Duplex / Villa Owner:**
"Dear Homeowner, every summer, your family faces a frustrating paradox: the hotter the sun shines outside, the more your air conditioners must run, pushing your meter into the highest penalizing tariff slabs. Why pay a utility company for power that arrives directly onto your rooftop for free? With an Ariel Rooftop System, the stronger the sun, the more free power your home produces. You insulate your family against tariff hikes, reduce bills to near zero, and teach your children what true environmental stewardship looks like in practice."

**For the Apartment Management Society:**
"Dear Flat Owners, every month, the common building service charge is a source of friction among neighbors — particularly the high cost of elevators, water pumps, corridor lighting, and generators. An Ariel Apartment Common-Load Solar installation

permanently solves this communal challenge. The monthly service charges for every family drop by 40% to 55%. There is zero roof drilling; our non-penetrative ballast mounts protect the roof waterproofing completely. It is an investment made once that rewards every family every single month for the next 25 years."